Ipratropium

Mechanism of action

- Ipratropium is a bronchodilator.
- Ipratropium is an anticholinergic agent with predominantly antimuscarinic activity. It antagonises (blocks) acetylcholine receptors, causing vagal inhibition resulting in bronchodilation.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Bronchospasm secondary to asthma or COPD.
- ✓ Prominent bronchospasm secondary to airway burns, smoke inhalation or chest infection.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy, but ipratropium should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 0.5 mg for adults and children once only.

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Administration

- Administer nebulised undiluted (in combination with 5 mg of salbutamol) using:
  - Air as the driving gas for COPD.
  - Oxygen as the driving gas for asthma.

Common adverse effects

- Tachycardia.
- Dry mouth.
- Blurred vision. This usually only occurs with repeated doses.

Usual onset of effect

- 2-5 minutes.

Usual duration of effect

- 6 hours.

Usual preparation

- Ampoule containing 0.5 mg in 2 ml.

Pharmacokinetics

- Only a small amount of the nebulised dose is absorbed, with most of the dose being nebulised to the atmosphere. The inhaled ipratropium is absorbed through the lungs and some is swallowed.
- Excretion is predominantly via the urine.
- Clearance is prolonged if the patient has significant kidney impairment, but this does not alter the initial (loading) dose.

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Common interactions

- None.

Additional information

- Ipratropium does not have a significant role in the treatment of bronchospasm as a result of smoke, toxic gas inhalation or chest infection. However, it may be administered if bronchospasm is prominent.
- Ipratropium has been reported to cause worsening of glaucoma, but only with frequent doses of nebulised ipratropium (with the nebulised drug contacting the eyes) in the setting of poorly controlled glaucoma. Glaucoma is not a caution when administering a single dose.

More Information

Version history

22 November 2022 1.0.1
Update: Aligned with asthma and COPD CPGs.